

Figure 3: A view of the components

1. At the top, we have the title for our application. It's always a good practice to name your application and show it on the screen as well. We have set the title property of Screen1 as the name of the application, so the top grey bar will be the application title or you can make it within the screen by providing a label.
2. Below that, we have a score card which is hidden in the beginning and will be loaded as the leader board at the end of the game.
3. In the middle, we have labels of the time and the moves. We will display the time taken and the steps count to solve the puzzle.
4. Beneath that, we have our actual play area. We have placed 16 image components on the canvas. Assign a number to the 15 numbers and keep one block empty.
5. At the end, the 'Shuffle the tiles' button will begin a new session

each time to randomly change the numbers displayed on the board.

Now, let's head towards the block editor to define the behaviour. Let's discuss the actual functionality that we are expecting from our application.

1. The board will be loaded with the numbers 1 to 16, in sequence. On pressing the 'Shuffle the tiles' button, numbers will be randomly distributed on the square board.
2. If there is an empty block adjacent to any block that is touched, then

this block should move to this empty position, and a blank should be created in the place from which it has moved. This is just a swapping of positions.

3. Each move will increase the move count.
4. Our clock timer will be running in the background and will count the seconds. Make sure to check the 'Timer always fires' checkbox in the clock properties.
5. Once the blocks are arranged in sequence, we need to display a message on the screen, and the move count and the time taken should be recorded to be presented on the leader board.

So let's move on and add these behaviours using the block editor. I hope you remember how to switch from designer to block editor. There is a button right above the *Properties* pane to switch between them.

Block editor blocks

I have already prepared the blocks for you. All you need to do is drag the relevant blocks from the left side palette and drop them on the viewer. Arrange the blocks the same way as shown in the image. I will explain what each does and how it is called.

As soon as Screen1 is loaded, certain variables to control events later on will be declared. In Figure 4, all the necessary variables are declared with their initial state.

Before we present anything to the user, we need to reset everything by keeping all the blocks at their default position and setting the lists. How the Screen1.Initialize block takes care of that is displayed in Figure 5.

The *Shuffle* button will re-arrange the blocks and start the game. Figure 6 shows the blocks for events when you click on the *Shuffle* button.

The 'tile_shuffle' procedure is a recursive call to shuffle the blocks; we have put that line of blocks within a procedure name so that we can call it any number of times.

The counter will be triggered as



Figure 4: Block editor image 1



Figure 5: Block editor image 2



Figure 6: Block editor image 3

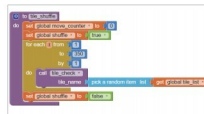


Figure 7: Block editor image 4